Advanced Empirical Finance: Topics and Data Science

The SDF, the efficient frontier and the CAPM

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From portfolios to prices

- Lectures 2 to 5: given expected returns, **which portfolio** should an investor hold?
- Now the other way round: given how investors choose, **which expected returns** do assets earn?
- Both questions meet in the same object: the efficient (tangency) portfolio

> **Today:** one equation that prices every asset, its link to the efficient frontier, and the CAPM as its best-known special case

The stochastic discount factor

Whiteboard: the fundamental equation

An investor with utility $u(c)$ can buy an asset at price p_t that pays x_{t+1} next period. When is she indifferent about buying a little more? (Cochrane, 2005, chapter 1)

$$p_t = E_t\left[m_{t+1}\, x_{t+1}\right] \qquad m_{t+1} = \beta\, \frac{u'(c_{t+1})}{u'(c_t)}$$

- For returns: $1 = E[mR]$; the risk-free rate is $R_f = 1/E[m]$; for excess returns: $0 = E[mR^e]$
- Expected excess returns come from the covariance with m:

$$E[R_i^e] = -R_f\, \mathrm{Cov}(m, R_i^e)$$

- An asset that pays off in bad times, when m is high, insures the investor and earns a low return

Whiteboard: the SDF and the efficient frontier

Which m prices N given assets, and how volatile must it be? (Cochrane, 2005, chapters 5 and 6)

- An SDF that is linear in returns, $m = a + b'R$, prices the assets if and only if the portfolio with weights proportional to b is **mean-variance efficient**: the tangency portfolio of lecture 2 is an SDF
- $|E[R^e]| \leq \frac{\sigma(m)}{E(m)}\,\sigma(R^e)$ for every excess return, so

$$\frac{\sigma(m)}{E(m)} \geq \max \text{ Sharpe ratio}$$

- This is the **Hansen-Jagannathan bound** (Hansen & Jagannathan, 1991): the higher the best attainable Sharpe ratio, the more volatile any SDF must be

The bound in the data

- The market and the ten Fama-French industry portfolios, 1960–2025: the market alone has a Sharpe ratio of 0.46 a year, the tangency portfolio of all eleven 0.67
- So any SDF that prices these assets has $\sigma(m)/E(m) \geq 0.67$ a year
- With $m = \beta\, u'(c_{t+1})/u'(c_t)$ and power utility, $\sigma(m) \approx \gamma\, \sigma(\Delta c)$. Consumption growth varies by about 1–2% a year, so γ must be implausibly large: the **equity premium puzzle** (Mehra & Prescott, 1985)

The CAPM

Whiteboard: the CAPM as an SDF

Suppose the market portfolio is mean-variance efficient. What does that imply for m, and for expected returns?

$$m = a - b\,R_m \qquad \Longleftrightarrow \qquad E[R_i^e] = \beta_i\,E[R_m^e], \qquad \beta_i = \frac{\mathrm{Cov}(R_i, R_m)}{\mathrm{Var}(R_m)}$$

- The **security market line**: only the covariance with the market earns a premium; idiosyncratic risk does not (Sharpe, 1964; Lintner, 1965)
- In a regression $R^e_{i,t} = \alpha_i + \beta_i R^e_{m,t} + \varepsilon_{i,t}$, the CAPM says $\alpha_i = 0$ for every asset

Do high-beta industries earn more?

Ten industries, 1960–2025: do the industries with high market betas have higher average excess returns?

- Yes clearly
- A little
- No relation
- The opposite

Betas and alphas of ten industries

R

```r
capm <- industries |>
  nest(data = -industry) |>
  mutate(
    fit = map(data, \(d) summary(lm(ret_excess ~ mkt_excess, data = d))$coefficients),
    alpha = map_dbl(fit, \(f) f["(Intercept)", "Estimate"]),
    t_alpha = map_dbl(fit, \(f) f["(Intercept)", "t value"]),
    beta = map_dbl(fit, \(f) f["mkt_excess", "Estimate"]),
    mean = map_dbl(data, \(d) mean(d$ret_excess))
  ) |>
  select(industry, alpha, t_alpha, beta, mean)
```

`industries` holds the monthly excess returns of the ten Fama-French industries, with the market excess return from `factors_ff3_monthly.parquet`

Betas and alphas of ten industries

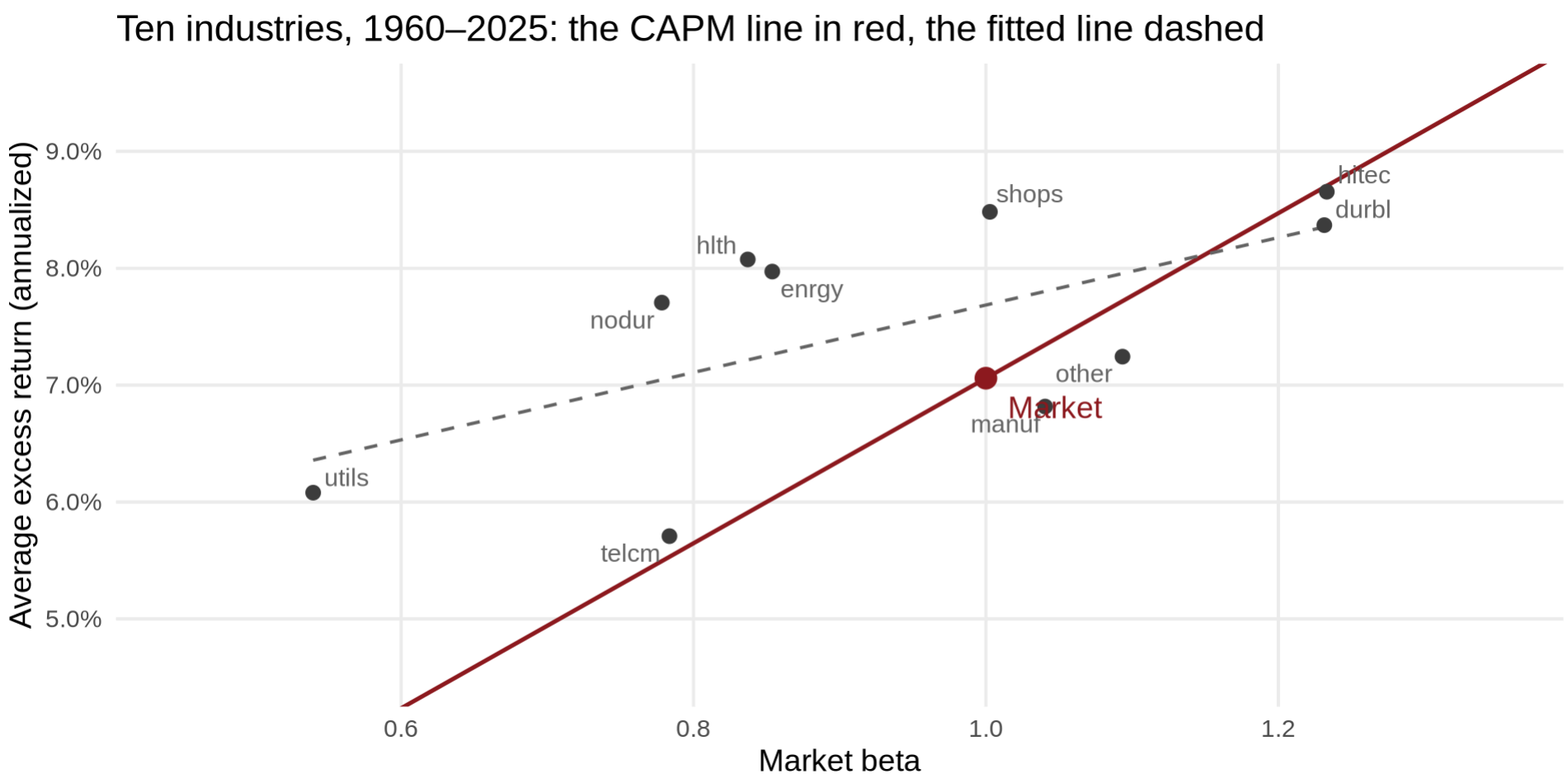
Python

```
factors_ff3_monthly = pl.read_parquet("data/factors_ff3_monthly.parquet")
industries = (
    pl.read_parquet("data/industries_ff_monthly.parquet")
    .unpivot(index="date", variable_name="industry", value_name="ret")
    .join(factors_ff3_monthly, on="date")
    .with_columns(ret_excess=pl.col("ret") - pl.col("risk_free"))
)

rows = []
for (industry,), data in industries.group_by("industry"):
    fit = smf.ols("ret_excess ~ mkt_excess", data=data.to_pandas()).fit()
    rows.append({"industry": industry, "alpha": fit.params["Intercept"],
                 "t_alpha": fit.tvalues["Intercept"], "beta": fit.params["mkt_excess"],
                 "mean": data["ret_excess"].mean()})
capm = pl.DataFrame(rows).sort("industry")
```

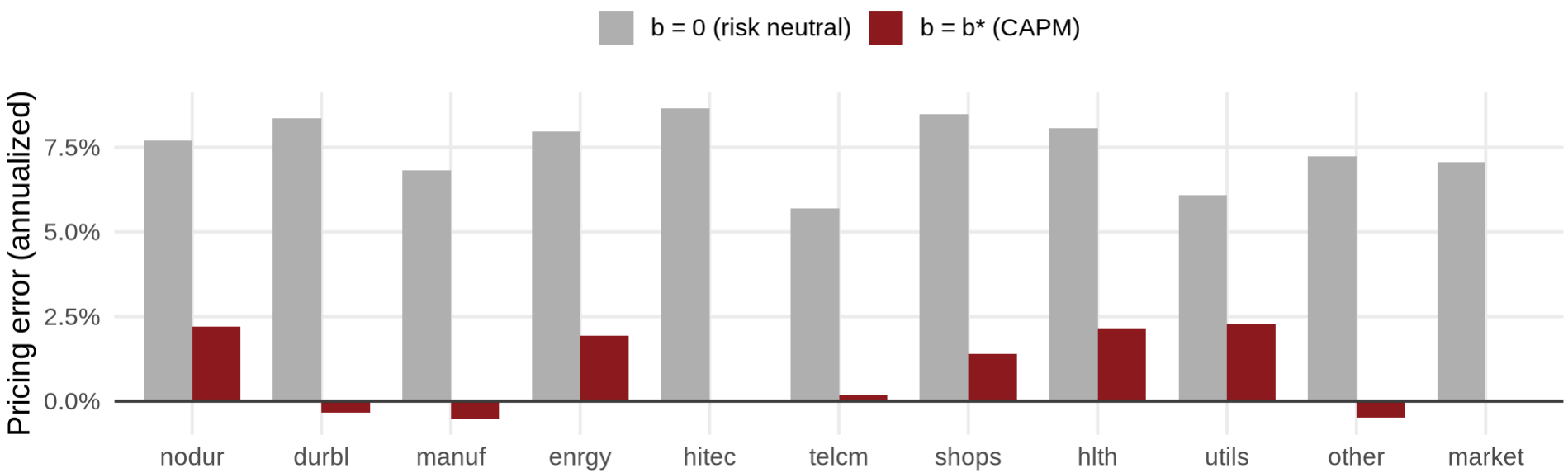
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The security market line



Pricing with the CAPM's SDF

- Take $m = 1 - b\left(R_m^e - E[R_m^e]\right)$. The pricing error of asset i is $E[mR_i^e] = E[R_i^e] - b\,\mathrm{Cov}(R_m^e, R_i^e)$
- $b = 0$: a risk-neutral investor, the errors are the average excess returns. With the CAPM's b^*, the market is priced exactly and the errors of the industries are their alphas



Testing the CAPM

Alphas

Industry	Beta	Average excess return	Alpha (annualized)	t-statistic of alpha
hitec	1.23	8.7%	-0.1%	-0.04
durbl	1.23	8.4%	-0.3%	-0.16
other	1.09	7.2%	-0.5%	-0.55
manuf	1.04	6.8%	-0.5%	-0.70
shops	1.00	8.5%	1.4%	1.30
enrgy	0.85	8.0%	1.9%	0.98
hlth	0.84	8.1%	2.2%	1.61
telcm	0.78	5.7%	0.2%	0.13
nodur	0.78	7.7%	2.2%	2.10
utils	0.54	6.1%	2.3%	1.63

- Low-beta industries tend to have positive alphas, high-beta industries negative ones: the security market line is flatter than the CAPM says

Whiteboard: the GRS test

Test all alphas at once, $H_0 : \alpha_1 = \ldots = \alpha_N = 0$, from T months of time-series regressions on the market

$$\text{GRS} = \frac{T - N - 1}{N} \frac{\hat{\alpha}' \hat{\Sigma}^{-1} \hat{\alpha}}{1 + \hat{\mu}_m^2 / \hat{\sigma}_m^2} \sim F(N, T - N - 1) \qquad \hat{\alpha}' \hat{\Sigma}^{-1} \hat{\alpha} = \widehat{SR}^2_{\text{market + assets}} - \widehat{SR}^2_{\text{market}}$$

- $\hat{\Sigma}$ is the covariance matrix of the residuals (Gibbons, Ross & Shanken, 1989)
- The test asks lecture 2's question: does adding the assets to the market raise the highest attainable Sharpe ratio?

GRS for the ten industries

R

```r
y <- industries_wide |> select(-date, -mkt_excess) |> as.matrix()
x <- industries_wide$mkt_excess
n_months <- nrow(y)
n_assets <- ncol(y)

fit <- lm(y ~ x)
alpha <- coef(fit)[1, ]
sigma_resid <- crossprod(residuals(fit)) / n_months
sharpe_market_2 <- mean(x)^2 / mean((x - mean(x))^2)

grs <- (n_months - n_assets - 1) / n_assets *
  as.numeric(t(alpha) %*% solve(sigma_resid, alpha)) / (1 + sharpe_market_2)
p_value <- 1 - pf(grs, n_assets, n_months - n_assets - 1)
```

GRS = 1.51, p-value 0.13: the ten industries do not reject the CAPM. The tangency portfolio raises the Sharpe ratio from 0.46 to 0.67, but over 792 months that is not enough evidence

GRS for the ten industries

Python

```
wide = (
    industries.pivot(on="industry", index="date", values="ret_excess")
    .join(factors_ff3_monthly.select("date", "mkt_excess"), on="date")
    .sort("date")
)
y = wide.drop("date", "mkt_excess").to_numpy()
x = wide["mkt_excess"].to_numpy()
n_months, n_assets = y.shape

design = np.column_stack([np.ones(n_months), x])
coef, *_ = np.linalg.lstsq(design, y, rcond=None)
alpha = coef[0]
residuals = y - design @ coef
sigma_resid = residuals.T @ residuals / n_months
sharpe_market_2 = x.mean() ** 2 / x.var()

grs = ((n_months - n_assets - 1) / n_assets
       * alpha @ np.linalg.solve(sigma_resid, alpha) / (1 + sharpe_market_2))
p_value = 1 - f_distribution.cdf(grs, n_assets, n_months - n_assets - 1)
```

GRS = 1.51, p-value 0.13: the ten industries do not reject the CAPM. The tangency portfolio raises the Sharpe ratio from 0.46 to 0.67, but over 792 months that is not enough evidence

What else is wrong with the CAPM?

- **Power:** industries are similar to each other and to the market. Portfolios sorted on size and book-to-market have large CAPM alphas (Fama & French, 1993). Next lecture: all stocks, sorted on beta
- **A flat security market line:** low-beta stocks earn too much, high-beta stocks too little (Fama & French, 2004; Frazzini & Pedersen, 2014)
- **The market is unobservable:** a test of the CAPM tests the proxy, here the CRSP value-weighted index (Roll, 1977)
- **Betas move over time:** exercise 4 estimates them with rolling windows for every CRSP stock